# Respiratory: Carbon Monoxide (CO) Poisoning and Cyanide (CN) Poisoning

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Adapted from: Carbon Monoxide Poisoning and Cyanide Poisoning

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Learning Objectives (5) Versions

*After completing this brick, you will be able to:*

1. Define carbon monoxide (CO) poisoning and cyanide (CN) poisoning and their sources.
2. Differentiate the clinical presentation of patients with CO poisoning or CN poisoning.
3. Explain the pathophysiology of CO and CN poisoning.
4. Explain how CO poisoning and CN poisoning are diagnosed.
5. Explain the management of CO and CN poisoning.

**CASE CONNECTION**

LB, a 40-year-old male is found comatose at the scene of a residential fire and brought to the emergency department. He presents with tachycardia, tachypnea, hypotension, and is unresponsive to painful stimuli. He has a singed beard, hair, and superficial burns on his left arm, and his skin is red in the unburned areas. Pulse oximetry shows normal oxygen saturation. He is immediately started on 100% oxygen, and his vitals improve. What is the most likely explanation of his condition, and what should be done next?

Consider your answers as you read, and we'll revisit LB at the end of the brick.

GO TO CONCLUSION ↓

## What Are Carbon Monoxide Poisoning and Cyanide Poisoning (LO1)?

Carbon monoxide (CO) and cyanide (CN) poisoning are life-threatening conditions that frequently occur together in smoke inhalation scenarios. Both toxins cause cellular hypoxia through different mechanisms, while patients may appear to have adequate oxygenation saturation by pulse oximeter monitoring. Severe poisoning can cause death if a quick diagnosis and treatment are not made.

- **CO poisoning** occurs when CO gas is inhaled and **reversibly** binds to hemoglobin, impairing oxygen delivery and causing tissue hypoxia.
- **CN poisoning** occurs when inhaled cyanide gas or ingested $CN^-$ ions **irreversibly** inhibits cellular respiration leading to profound tissue hypoxia and severe lactic acidosis.

## Sources and Exposure

CO is a colorless, odorless gas that blocks oxygen delivery, while CN—commonly inhaled as hydrogen cyanide or absorbed as salts—is rapidly acting and highly lethal. Awareness of these sources is essential for early recognition and treatment.

**Carbon monoxide (CO)** sources include:

- Incomplete combustion of carbon-containing fuels (wood, gas, coal, propane)
- House fires, especially in enclosed spaces
- Faulty heating systems, furnaces, and water heaters
- Industrial exposure from blast furnaces and coke ovens
- Methylene chloride (paint removers, solvents) is metabolized to CO in the liver

**Cyanide (HCN)** sources include:

- Combustion of synthetic materials (plastics, polyurethane, nylon, silk) in fires
- Industrial processes using cyanide compounds (mining, electroplating)
- Iatrogenic: Sodium nitroprusside metabolism (IV vasodilator medication used for hypertensive emergencies) because the drug is metabolized to CN

- Ingestion of cyanogenic plants and foods (apricot kernels, bitter almonds, cassava)
- Laboratory chemicals and pest control substances that contain CN

Inhaled fumes during fires is the most common source of both CO and CN poisoning.

## Epidemiology

Smoke inhalation accounts for 75–80% of fire-related deaths in the United States. Carbon monoxide (CO) poisoning affects about 50,000 people annually, with a 1–3% mortality rate, most often in winter with increased heater use; 57% of deaths are linked to automobile exhaust. While exact data on cyanide (CN) are limited, an estimated 35–40% of fire victims have toxic CN levels, though overt symptoms of CN poisoning are rare.

# Clinical presentation of Carbon Monoxide or Cyanide Poisoning (LO2)

## Carbon Monoxide Poisoning

Clinical symptoms of CO poisoning are variable and nonspecific, and vary with exposure. Generally, there is slow onset alteration of mental status, and it is often misdiagnosed as viral syndromes. The most common symptoms are listed below (Manaker and Perry 2019):

- Headache
- Nausea
- Dizziness
- Drowziness
- Vomiting

**Severe toxicity/ exposure:**

- *Central nervous system (CNS):* seizures, syncope, or coma
- Delayed neuropsychiatric syndrome sequelae can develop days to weeks after survival of exposure: cognitive deficits, personality changes, and movement disorders.
- *Metabolic:* mild increases in lactic acidosis (3 mM).
- *Cardiovascular:* acute myocardial ischemia, ventricular arrhythmias, and pulmonary edema.

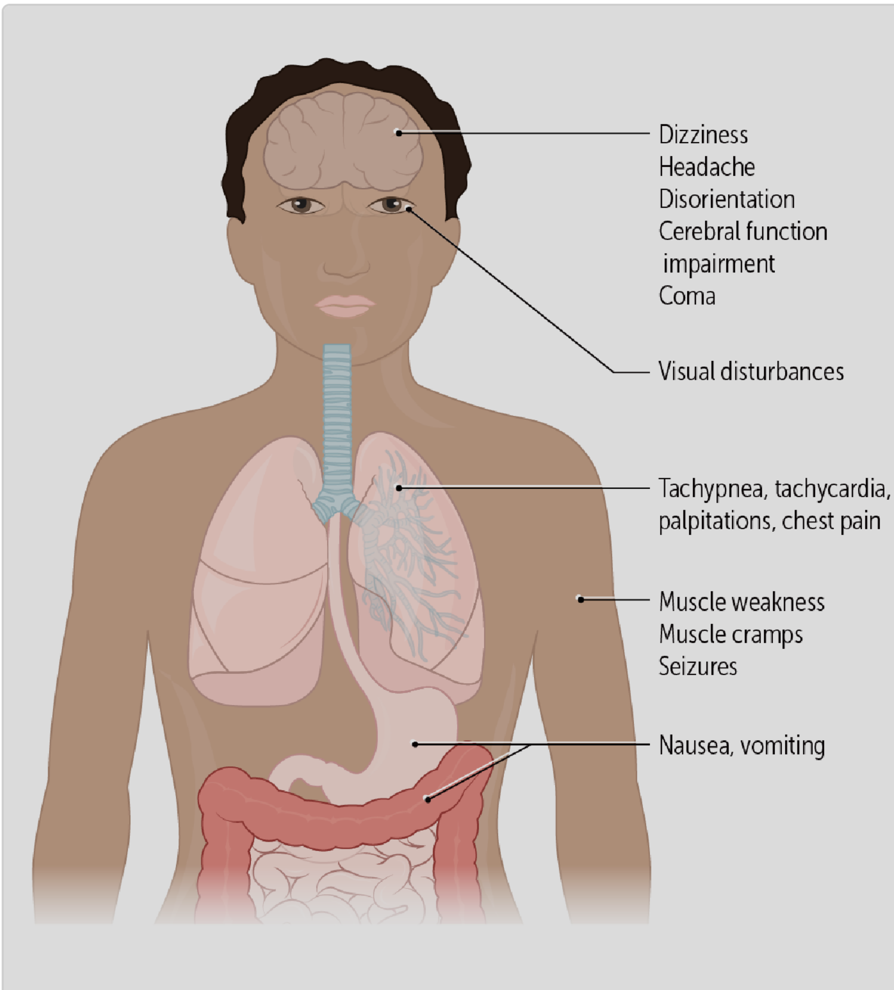


**Figure 1. CO symptoms**

**CLINICAL CORRELATION**

Neurologic symptoms in carbon monoxide (CO) poisoning can easily be overlooked, since many patients complain of them, and most are caused by more common conditions. That's why CO poisoning should always be on your mind as a clinician.

## Cyanide Poisoning

Symptoms of cyanide poisoning depend on the toxicity level, route (inhaled or ingested), with central nervous system (CNS) and cardiovascular changes being the most common (Desai 2019). Some patients, particularly after inhaling hydrogen cyanide gas, may also report a characteristic "bitter almond" odor or taste.

***Acute toxicity/ exposure***

- *Central nervous system (CNS):* headache, anxiety, confusion, dizziness, syncope and seizures. A noted and diagnostic symptom is mydriasis (pupil dilation).
- Delayed neuropsychiatric syndrome sequelae can develop days to weeks after survival of exposure: delayed onset Parkinson's.
- *Cardiovascular:* initial tachycardia and hypertension, followed by bradycardia, hypotension, and dysrhythmias.
- *Metabolic:* **High lactic acidosis (> 8 mM)**
- *Respiratory:* Initial tachypnea, then bradypnea and pulmonary edema
- *Gastrointestinal:* abdominal pain, vomiting, hepatic necrosis
- *Skin:* The cheery red skin color is an unreliable sign, and occurs in only in a few cases. It may be seen in retinal veins (bright retinal veins).
- *Renal:* kidney failure

***Chronic toxicity/ exposure***

- tends to be associated with vague symptoms: headache, vomiting, chest and abdominal pain, and anxiety.

## What Is the Pathophysiology of Carbon Monoxide Poisoning (LO3)?

The pathophysiology of CO poisoning is related to the **strong affinity that CO has for heme (240 times greater than $O_2$)**, a component of hemoglobin (Hb) and myoglobin molecules. Because CO binds the same hemoglobin site as $O_2$, CO can outcompete $O_2$ for the binding site. Additionally, CO also increases hemoglobin's affinity for any remaining bound oxygen, preventing $O_2$ release to tissues. The result is functional anemia and tissue hypoxia due to poor $O_2$ delivery by the RBC. Comparing the same 50% reduction of hemoglobin O2 carrying capacity for anemia (reduced hemoglobin levels) and CO poisoning, CO poisoning results in greater hypoxia due to the combination of both increased affinity and competitive displacement of $O_2$. In addition, CO impairs cellular respiration, though less severely than cyanide.

### Effects of CO on Hemoglobin

This abnormal hemoglobin binding effect can be shown on the oxygen-hemoglobin dissociation curve, which shifts to the left with CO exposure, resulting in decreased oxygen release to tissues at a given level of blood $O_2$ concentration (Figure 2). In Figure 2, the p50 is the pressure at which hemoglobin is 50% saturated, and it is shifted to the left for CO bound to hemoglobin (carboxyhemoglobin, COHb).

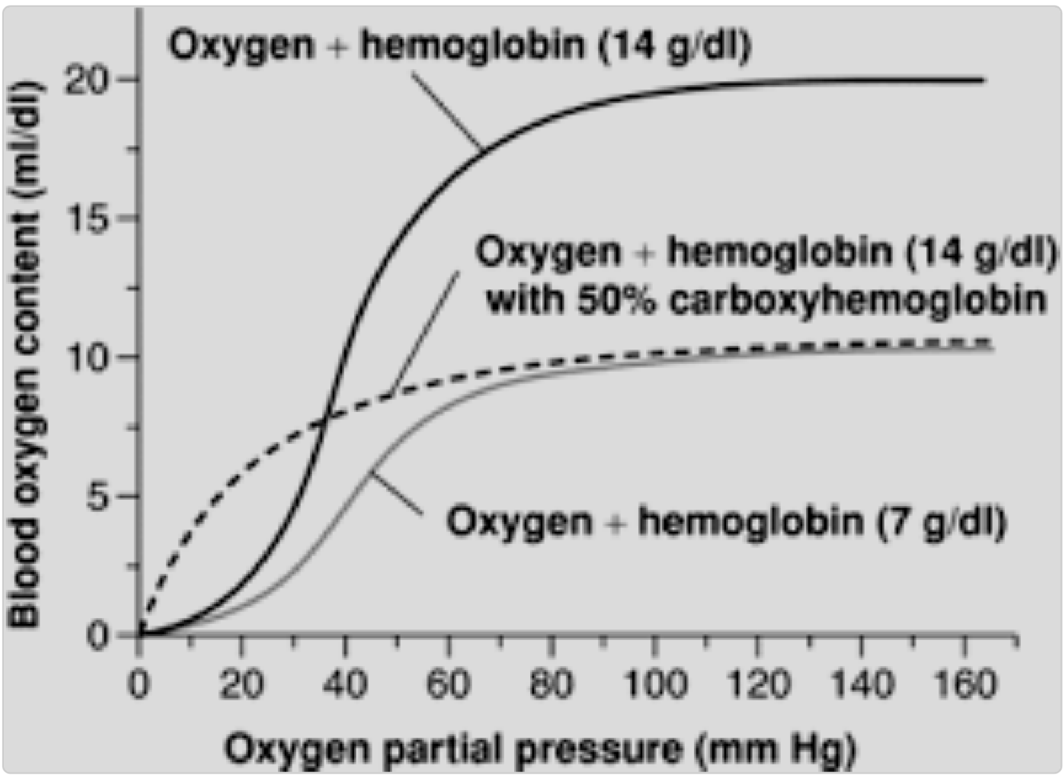


**Figure 2. Comparison of 50% reduction $O_2$ carrying capacity for anemia and CO. CO results in an increased O2 binding to hemoglobin (left shift), and decreased carrying capacity (content) vs 50% anemia**

## Effects of CO on Myoglobin

CO binding also affects myoglobin, another heme-containing protein mostly present in muscles (including cardiac muscle) which is important in $O_2$ delivery to muscles. CO binds to myocardial myoglobin, preventing $O_2$

release, resulting in myocardial ischemia. This can lead to angina, myocardial infarction (MI), and heart failure with hypotension or shock.

Carboxyhemoglobin has a high affinity for oxygen, so it does not release oxygen molecules even at low partial pressures of oxygen. This causes the leftward shift in the oxygen-hemoglobin dissociation curve.

## Systemic Effects of CO

In CO poisoning and the resulting tissue hypoxia, anaerobic respiration occurs, leading to **lactic acidosis**. Inhibition of the enzyme cytochrome C oxidase (which is key to oxidative phosphorylation) leads to the inability of cells to use $O_2$, with eventual cell death (necrosis).

Some specific cellular toxicities and effects include:

- *Neurologic system:* CO binding to hemoglobin shifts the oxygen dissociation curve left, reducing oxygen delivery to the brain. The resulting hypoxia increases cerebral blood flow, causing cerebral edema and symptoms such as headache, dizziness, weakness, nausea, and confusion.

- *Respiratory system:* Direct toxicity to alveolar membranes leads to pulmonary edema, reduced oxygen diffusion, tachypnea, and, in severe cases, respiratory failure.
- *Cardiovascular system:* CO binding to myocardial myoglobin impairs oxygen release, causing ischemia, chest pain, arrhythmias, and potentially infarction or heart failure. Heart failure can further worsen pulmonary edema. Arrhythmias may include atrial fibrillation, prolonged QT, and ventricular tachyarrhythmias.
- *Metabolic:* Hypoxia results from impaired oxygen delivery. 10–15% CO also binds myoglobin, cytochrome, and NADPH reductase, inhibiting mitochondrial oxidative phosphorylation and leading to anaerobic metabolism with lactic acidosis.
- *Skin (unreliable):* A cherry-red discoloration of skin or lips (in persons with lighter complexion) may occur in severe poisoning. However, this finding is neither sensitive nor specific and is now uncommon in living patients. It is a cardinal feature in forensic pathology and autopsy evaluation rather than clinical presentation.

## What Is the Pathophysiology of Cyanide Poisoning (LO3)?

Like CO poisoning, CN poisoning also leads to cell necrosis but by inhibiting key intracellular pathways. It is much less common than CO poisoning.

### Cell Biology

Cyanide (CN) toxicity results from **inhibition of oxidative phosphorylation**. CN binds to ferric iron ($Fe^{3+}$) in mitochondrial cytochrome C oxidase, blocking the electron transport chain and preventing oxygen use.

This causes cellular hypoxia, halts ATP production, and forces anaerobic metabolism, leading to lactic acidosis and cell death.

## Systemic Effects

**Neurologic:** mental state alteration occurs because of cerebral edema. Other neurologic effects include optic neuropathy (vision loss) and Parkinsonian symptoms (eg, ataxia) from basal ganglia injury. A noted and diagnostic symptom is mydriasis. The cellular hypoxia leads to damage to the basal ganglia, but this is usually seen postmortem.

Other cellular effects include:

- *Cardiovascular system:* CO poisoning initially causes hypertension due to vasoconstriction. However, this is followed by bradycardia and hypotension with subsequent chest pain from arrhythmias due to myocardial injury.
- *Gastrointestinal system:* hepatic failure occurs due to hepatocyte necrosis.
- *Renal system:* acute kidney injury may result from rhabdomyolysis, followed by myoglobin filtration into the tubule. There, the myoglobin is toxic to the renal tubules.
- *Respiratory system:* pulmonary edema may cause dyspnea and respiratory failure.
- *Metabolic*: Forces cells into anaerobic metabolism, causing s**evere lactic acidosis (> 8 mM)**.

# How Do We Diagnose Carbon Monoxide Poisoning and Cyanide Poisoning (LO4)?

Diagnosing carbon monoxide (CO) and cyanide (CN) poisoning is challenging. Both often begin with vague, nonspecific neurologic symptoms (eg, headache, dizziness) but can progress to severe, life-threatening illness. Both target the heart and brain, which are especially vulnerable to hypoxia.

A key difference is the timeline: CO poisoning typically evolves gradually over hours, whereas CN poisoning can cause rapid collapse within minutes, reflecting its immediate disruption of cellular metabolism. You should always keep these poisonings in mind, especially CO poisoning in winter months when heater use increases. In fire-related combined exposures, the clinical course may be mixed and rapidly progressive, making early recognition and intervention critical.

## Detecting Carboxyhemoglobin

The key to diagnosing CO poisoning is detecting **carboxyhemoglobin** (COHb) using blood gas analysis with a specific request or by CO-oximetry (Figure 3). A COHb level above 3% raises concern, though results can be misleading in smokers, who may have baseline levels up to 10%.

**Remember** that routine pulse oximetry is unreliable, as it cannot distinguish oxyhemoglobin from COHb and may falsely appear normal despite significant poisoning. Newer multiwavelength pulse oximeters (CO-oximeters) can detect COHb and provide a more accurate assessment, but these are not universally available in all clinical settings. The discrepancy between a normal pulse oximetry reading and clinical signs of hypoxia should raise suspicion for CO poisoning. Serum and blood values for CO poisoning are shown in Table 1, where $SpO_2$ denotes oxygen saturation measured by a pulse oximeter, and $SaO_2$, which is arterial blood gas saturation measured via an arterial blood gas (ABG) test.


Figure 3

**Table 1. Serum/ Blood measurements for CO**

| Parameter | Arterial Blood | Venous Blood gas |
|---|---|---|
| pH | 7.38 (normal to slightly acidic) | Normal |
| $PaO_2$ | 98 (normal) mmHg | 75 mmHg |
| $PaCO_2$ | 35-45 mmHg | 40 mmHg |

| Parameter | Arterial Blood | Venous Blood gas |
|---|---|---|
| $HCO_3^-$ | 24 mEq/L (normal or slightly reduced) | 24-28 mEq/L |
| $SpO_2$ | Falsely normal (pulse ox $O_2$ saturation) | -- |
| $SaO_2$ | arterial $O_2$ saturation varies with the CO saturation of Hb. Textbook examples usually provide examples for 50% | depends on CO saturation of Hb. |
| COHb | 25% (severe poisoning) | -- |
| Lactate | Elevated (3 mmol/L) | |
| A-a Gradient | Normal (5-10) | -- |

Pulse oximetry cannot differentiate between carboxyhemoglobin and oxyhemoglobin; therefore, it shows normal results even with the low oxyhemoglobin in CO poisoning.

## Detecting Cyanide

Unlike carbon monoxide (CO) poisoning, **blood cyanide (CN) levels are not helpful** for initial diagnosis because results can take hours to return and are mainly used for confirmation.

**Blood gas analysis** should be obtained though values for $PaO_2$, $PaCO_2$, and pH are usually normal, as CN does not affect the solubility or gas exchange, unless there is concurrent lung damage (e.g., from heat exposure during a fire). **Pulse oximetry** is unreliable since the $PaO_2$ is normal.

**Serum studies** provide additional clues: lactate is often elevated in both CO and CN poisoning, since lactic acid is released from necrosing cells. This is a **sensitive test for CN poisoning**, and a markedly elevated level (>10 mmol/L) in a fire victim suggests CN poisoning. The level of lactic acid correlates well with the degree of poisoning. The increased lactate causes an increase in the serum anion gap (see Table 2).

**Cardiac enzymes** (eg, troponin) released during myocardial cell necrosis should be measured in patients with chest pain, arrhythmias, or signs of heart failure. An ECG is also needed to detect myocardial injury.

**Table 2. Serum/ Blood measurements for CN**

| Parameter | Arterial Blood Gas | Venous Blood Gas |
|---|---|---|
| pH | 7.20 (acidotic) | 7.20 (acidotic) |
| $PaO_2$ | 80-100 (normal) | 84 mmHg (high; normal 40) |
| $PaCO_2$ | 39 mmHg (normal/low) | 44 mmHg (normal) |
| $HCO_3^-$ | 16 mEq/L (low) | |

| Parameter | Arterial Blood Gas | Venous Blood Gas |
|---|---|---|
| $SpO_2$ | 100% (normal, pulseox $O_2$ saturation) | 90% (high) (normal 75%) (A-V difference small, less $O_2$ extraction) |
| $SaO_2$ | arterial $O_2$ saturation (appears normal) | as above |
| CaO2 | normal | |
| Lactate | Elevated (> 8-10 mmol/L) | |
| A-a Gradient | Normal (5-10) | -- |

### Imaging

**Chest radiography** is done in symptomatic patients to evaluate for pulmonary edema, a potential consequence of heart failure induced by CO or CN poisoning. Patients with altered mental state or neurologic deficits may receive **brain CT** or **MRI**, looking for other causes of neurologic deficits (eg, trauma, strokes).

## How Do We Manage Carbon Monoxide Poisoning and Cyanide Poisoning (LO5)?

The first and the most important step in treating any poisoning is separating the patient from the poisoning source (eg, early removal from the fire site, stopping hospital administration of sodium nitroprusside).

## Supportive Therapy

Airway, breathing, and circulation are maintained with $O_2$ and, if required, intubation and rapid IV boluses of isotonic fluids and vasopressors. Seizures can be treated with benzodiazepines.

## Carbon Monoxide Poisoning

The most important initial treatment is **immediate administration of 100% oxygen**, delivered by nonrebreather mask or by intubation and mechanical ventilation if the patient is unconscious. Although CO binds hemoglobin with far greater affinity than oxygen, it can still be displaced when oxygen is present in overwhelming excess. High oxygen levels promote CO dissociation from hemoglobin, allowing it to be exhaled.

In severe cases, **hyperbaric oxygen therapy** is used to further increase arterial oxygen pressure and accelerate CO clearance. Indications for hyperbaric therapy include:

- COHb levels above 25%
- Pregnancy with COHb > 15%
- Loss of consciousness or altered mental status
- Evidence of tissue necrosis (such as myocardial infarction, respiratory failure or neurologic injury), or severe metabolic acidosis with blood pH below 7.25
- Blood pH <7.25 (severe metabolic acidosis, due to lactic acidosis)

## Cyanide Poisoning

Because CN poisoning can be rapidly fatal, antidotes should be given early in suspected poisoning, before CN drug levels return. **Hydroxocobalamin** is the drug of choice for both CN ingestion and inhalation. It binds to CN, forming cyanocobalamin, which is then excreted in the urine.

Suspected oral cyanide poisonings via ingestion (unusual) can be given activated charcoal in the stomach. This is mainly to bind the CN in the intestines and reduce further GI absorption. One would still administer hydroxocobalamin.

## Differentiating between HCN and CO poisoning

**Table**

| Cyanide | Carbon monoxide |
| --- | --- |
| Rapid onset; CNS, cardiovascular dysfunction, and coma | Slower, insidious onset |
| Mydriasis (dilated pupils) | No mydriasis |
| Very high lactate levels ( > 8-10 mM) | Elevated lactate but not as high as cyanide lactate |
| Dyspnea and acute hypoxemic respiratory failure. Cyanide poisoning initially have hypernea but will become apneic within 3-5 minutes | Dyspnea and acute hypoxemic respiratory failure |
| Diminished arterial–venous $O_2$ difference is mainly due to high venous $O_2$ levels. This is due to the tissue's inability to use oxygen at the mitochondria level, specifically binding to cytochrome C oxidase, thus preventing the cells from using oxygen. Providing supplemental oxygen with 100% $FiO_2$ is still recommended in cyanide poisoning, especially milder exposure, but may not work with severe poisoning. | Normal a-v $O_2$ difference (Normal $O_2$ use at tissues). Provide high flow supplemental $O_2$, and hyperbaric oxygen therapy |
| Initial tachycardia, and then bradycardia, leading to hypotension (cardiogenic stroke). Causes | Slower time course/ dose dependent |

| Cyanide | Carbon monoxide |
|---|---|
| sudden cardiovascular collapse | |
| Does not respond to supplemental $O_2$ | Responds to supplemental $O_2$ |

**CASE CONNECTION**

BACK TO INTRODUCTION

Thinking back to LB, what accounts for his condition? LB's exposure to a house fire puts him at risk for both CO and CN toxicity; the CO would explain his red color. Note the red skin color should not be used in diagnosis, as it is infrequent, and may not be readily observed in patients with darker skin color. Neither condition would show hypoxemia on pulse oximetry. CO-oximetry would be done next to reveal if LB's carboxyhemoglobin level is elevated; serum lactic acid levels and arterial blood gases should be measured, and ECG and chest radiography should be done to look for other possible complications. He should be continued on 100% oxygen and hyperbaric oxygen, considered if his CO ingestion is severe or if evidence of end-organ damage is seen (e.g., myocardial infarction, heart failure, or pulmonary edema). If the lactate levels are high, there may be some CN poisoning in addition to CO poisoning. Thus, consider using an antidote to CN poisoning, like hydroxocobalamin, until the blood CN levels return.

# Summary

- Carbon monoxide (CO) poisoning and cyanide (CN) poisoning are often consequences of home fires.
- CO is given off as exhaust fumes from fires, vehicles, and heaters, as it is the byproduct of incomplete combustion.

- CN is given off as a byproduct of the combustion of rubber and plastic.
- CN poisoning can occur after industrial exposure, by ingesting chemicals, or after prolonged use of sodium nitroprusside for hypertension.
- Both CO and CN poisoning most often initially present with nonspecific neurologic symptoms like headache, dizziness, weakness, nausea, and confusion.
- CO shifts the oxygen-hemoglobin dissociation curve to the left.
- CO reversibly and competitively combines with hemoglobin with higher affinity than oxygen, forming carboxyhemoglobin (COHb), which decreases the oxygen-carrying capacity of blood; therefore, oxygen is not released to the tissues, causing reduced $O_2$ release from Hb, and tissue hypoxia.
- Tissue hypoxia can lead to respiratory failure, heart failure, myocardial infarction, and cerebral edema with neurologic deficits.
- CN inhibits cytochrome oxidase C, so oxidative phosphorylation cannot take place, oxygen in the cell is not used, and cellular hypoxia and necrosis result.
- Anaerobic respiration leads to severe elevations in lactic acid. Lactic acid elevation is a sensitive marker for CN poisoning (usually >8 mM).
- In CO poisoning, COHb can be detected by CO-oximetry or on blood gas analysis. However, the oxygen saturation appears normal on routine pulse oximetry.
- In CN poisoning, CN blood levels are diagnostic but often return too late for initial treatment.
- For CO poisoning, use 100% oxygen, delivered in a hyperbaric chamber in severe poisoning.
- For CN poisoning, use the antidote hydroxocobalamin.

# Review Questions

1. Four people are admitted to the emergency department after being rescued from a car that was stuck in a snowstorm for more than 20 hours. The vehicle's heater was running the whole time to keep them warm. The tailpipe was clogged with snow. Three of the patients are dizzy, confused, and have headaches. One is comatose. Pulse oximetry shows oxygen saturation of 97%. The chest x-rays are normal. Which of the following is the most likely diagnosis?

A. Acute respiratory distress syndrome

B. Carbon monoxide poisoning

C. Congestive heart failure

D. Cyanide poisoning

E. Pulmonary edema

**Explanation** (requires correct answer)

**2.** Three roommates are admitted to the emergency department with confusion and dizziness. They had been using an old wood-burning furnace, and all the windows and doors of their home had been shut. They have no past medical history. They do not have histories of smoking, drinking, or drug abuse. Their vital signs are stable, and oximetry oxygen saturation is normal. Which of the following is the most appropriate first step in management of these patients?

**A.** 100% oxygen

**B.** Activated charcoal

**C.** Hydroxocobalamin

**D.** Hyperbaric oxygen

**Explanation** (requires correct answer)

oxygen-dissociation curve for this patient?

**A.** Left due to CN

**B.** Left due to CO

**C.** Right due to CN

**D.** Right due to CO

**Explanation** **(requires correct answer)**

**4.** A 45-year-old female presents with confusion and dyspnea after house fire exposure. Which lab finding would most likely indicate combined CO and cyanide poisoning?

**A.** Elevated blood lactate

**B.** Elevated $PaO_2$

**C.** Low COHb level

**D.** Normal pulse oximetery

**Explanation** (requires correct answer)

**5.** A firefighter developed seizures and dilated pupils following a prolonged fire exposure. Which treatment is appropriate for suspected cyanide poisoning?

**A.** 100% oxygen

**B.** Activated charcoal

**C.** Hydroxocobalamin

**D.** Intravenous fluids

**Explanation** (requires correct answer)

**A.**

**B.** Binding to myocardial myoglobin

**C.** Direct cytotoxicity of CO on cardiac cells

**D.** Enhanced tissue oxygenation

**E.** Increase in alveolar oxygen exchange

**Explanation** (requires correct answer)

## How would you rate this Brick?

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## Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 **Define carbon monoxide (CO) poisoning and cyanide (CN) poisoning and their sources.**

02 Differentiate the clinical presentation of patients with CO poisoning or CN poisoning.

03 Explain the pathophysiology of CO and CN poisoning.

04 Explain how CO poisoning and CN poisoning are diagnosed.

05 Explain the management of CO and CN poisoning.

**Objective Confidence: 0/5**